

EdgeSign: An IoT-Enabled Edge–Cloud Hybrid System for Real-Time Sign Language Translation

Surya Sathyanarayanan
*Department of Computational
Intelligence*
S.R.M Institute of Science and
Technology
Chennai, India
ss7626@srmist.edu.in

Jeevan Babu Murugan
*Department of Computational
Intelligence*
S.R.M Institute of Science and
Technology
Chennai, India
jm7531@srmist.edu.in

Dr. R Babu
*Department of Computational
Intelligence*
S.R.M Institute of Science and
Technology
Chennai, India
babur@srmist.edu.in

Abstract—Real time gesture to speech systems play a crucial role in improving communication accessibility for individuals with speech and hearing impairments. The proposed assistive communication framework is implemented using Raspberry Pi 4 along with the Raspberry Pi Camera Module 3 to capture continuous hand gesture movements. MediaPipe Hands is utilized to extract twenty one hand landmarks, which are further processed to classify gestures with high accuracy. Unlike conventional approaches that translate gestures into isolated words, the system integrates an AI driven sentence generation mechanism to construct meaningful and grammatically coherent sentences from gesture sequences. Text to speech synthesis is performed on the Raspberry Pi to produce audible communication output. The system achieves low latency while maintaining portability and cost efficiency, making it suitable for educational, clinical, and social environments. Experimental evaluation confirms the reliability of the gesture recognition pipeline and demonstrates the system's capability to function as a standalone assistive communication solution.

Keywords—Gesture Recognition, Raspberry Pi, MediaPipe, Assistive Technology, Real Time Processing, AI-Driven Sentence Generation, Computer Vision.

I. INTRODUCTION

Communication plays an important role in social interaction, education, and professional life. For individuals with speech or hearing impairments, conveying information to people who are not aware of sign language often becomes difficult. In many real world situations, this gap leads to misunderstandings, delays, or complete communication breakdowns. While sign language is a useful and expressive medium, it is limited among the people and restricts seamless interaction in public and institutional environments.

Various technological solutions have been found to address this challenge, which ranges from sensor based gloves to camera driven recognition systems. But many of these solutions are costly, hardware intensive, or not suitable for portable deployment. Recent progress in embedded vision frameworks and compact computing platforms has opened new possibilities for developing lightweight, camera based gesture recognition systems. Real time hand landmark detection is used along with optimized computer vision pipelines which create accurate gesture understanding without the help of specialized sensors.

The goal of this effort is to create a gesture to speech communication system that is small, effective, and useful that runs on inexpensive technology. The system attempts to provide a natural and practical communication tool by integrating intelligent sentence synthesis, gesture sequence interpretation, and real time landmark extraction. Instead of being restricted to controlled conditions, the suggested approach is made to be effective in ordinary communication by placing a high priority on low reaction time.

II. LITERATURE REVIEW

To record hand movements, early sign language recognition systems mostly used sensor based techniques like data gloves and wearable sensors. These techniques were costly, difficult to use for an extended period, and required frequent updates in the values, which created difficulties in their practical acceptance even though they produced precise results. Researchers turned to vision based methods with cameras to get around these problems and enable non intrusive gesture identification.

Camera based gesture recognition systems that use hand tracking frameworks and deep learning models are the result of recent developments in computer vision and machine learning. Even though these techniques increased the accuracy of recognition, many of them are unsuitable for low power embedded platforms and need a lot of processing power. Efficient landmark extraction was made possible by lightweight frameworks like MediaPipe, which allowed for real time gesture recognition on gadgets like the Raspberry Pi. There is a need for more expressive and effective assistive communication solutions because most of the current systems only recognize individual gestures and lack contextual comprehension or sentence level interpretation.

III. MOTIVATION

Despite significant advancements in computer vision and artificial intelligence, practical sign language translation systems remain limited in real world deployment. Many existing solutions require high computational resources, continuous cloud connectivity, or controlled environments, making them unsuitable for everyday use. There is a growing need for a portable, low cost, and real time assistive system that can operate efficiently on edge devices while still leveraging cloud intelligence when required.

The motivation for this work arises from the need to develop an affordable, portable and real-time assistive communication system that can be used in everyday environments. Recent developments in embedded computing and computer vision have enabled the design of lightweight, camera-based gesture recognition systems on affordable hardware such as the Raspberry Pi. By combining efficient hand landmark detection, gesture interpretation, and speech generation this project seeks to deliver a practical solution that improves accessibility, fosters independence and helps to make natural communication for sign language users.

This work creates a growing need for assistive technologies that translate gestures and support natural, context-aware communication. Many current systems overlook sentence creation, conversational flow, and real-time feedback, all of which are essential for effective interaction. The goal is to connect this gap by merging gesture recognition with smart sentence production and multiple output. This ensures that communication is clear and expressive. The focus is on practical use, low power consumption, and protecting privacy, making the system suitable for everyday use instead of controlled or lab settings.

IV. OBJECTIVES

The purpose of this undertaking is to develop a real-time interactive communications system to facilitate communication between people who use American Sign Language (ASL) and those who do not. A video camera will capture the user's hand gestures and convert them into meaningful spoken and textual responses. Embedded hardware with modest cost will be used for processing. The focus of this development will be to obtain an accurate recognition (what occurred) with an extremely short latency period (how long it takes for data to be completely processed), creating a natural communication environment as if both parties were communicating in their everyday lives.

The other goal of the research is to recognize and support both static and dynamic gestures using suitable data representation techniques and modeling approach. Static gestures are instantly recognized using the feature points based upon landmark features and dynamic gestures is expressed as a collection of landmarks extracted from all the frame to capture the motion characteristics. To acquire this a specific gesture data is collected using Raspberry Pi camera with close likeness to how the application is used in practice.

Fig. 1 shows how the gesture samples in the dataset are stored as .npy files, which contains extracted hand landmark data from MediaPipe. The gestures we have categorized for this dataset are call, eat, go, hello, help. The dataset is organized very systematically into class folders, further divided into two sets- folders named train and test. The folder organization of dataset is mainly to help in training and evaluation in structured way while training the model, and also helps to avoid some common errors that arise during model training. By evaluating the model on the test data, the reliability of our model can also be predicted and the ability to generalize is also tested. If the model is able to

give the desired output on the test data which consists of different set of data than the data on which it was trained on, it can be said that the model is a good one. Hence our dataset is divided into train and test. The dataset is specially created to be used in EdgeSign framework, which is an easy to use and lightweight framework to implement deep learning models on Edge devices.

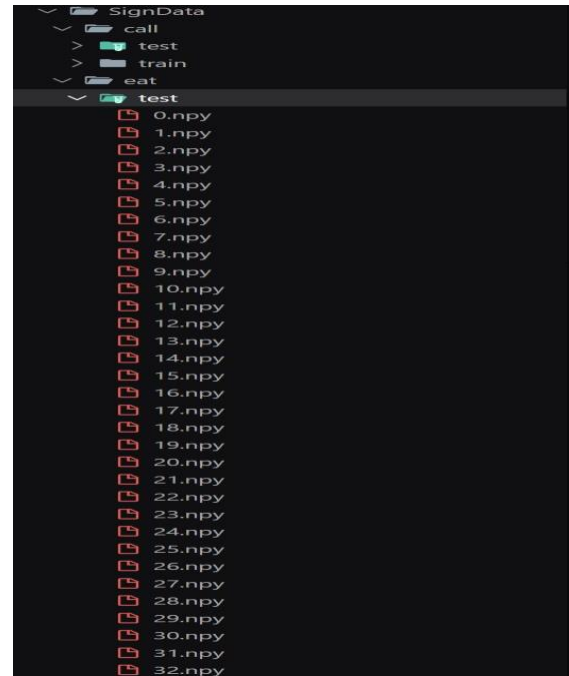


Fig. 1. Image of Dataset

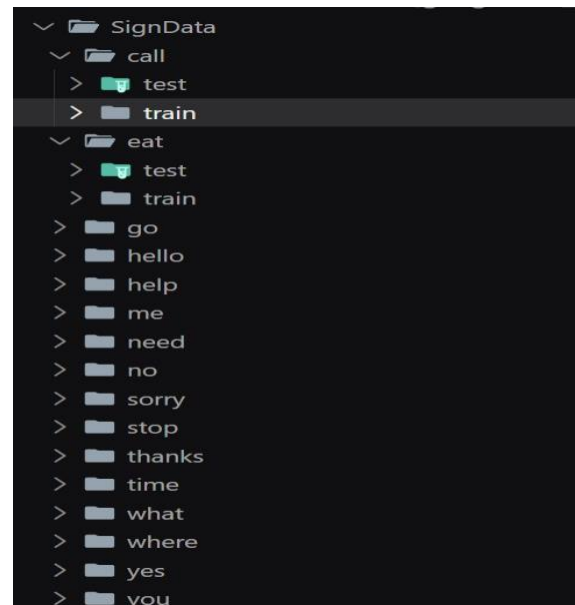


Fig. 2. Another labels of Dataset

The system includes a way to facilitate reliable communication through its efficient coordination of edge and cloud. By performing hand landmark extraction via a Raspberry Pi and communicating only structured landmark data rather than unstructured raw video frames, through this approach it will be able to save network traffic and protect user privacy. Additionally, this design choice will allow for a quicker response time to fulfill use cases in locations where

only limited connections are available. We want to achieve a balanced architecture in which computationally intensive jobs are done remotely when required, while still being able to maintain real-time performance at the edge.

An additional key goal is to improve usability by adding a variety of modalities of feedback and contextual information. A compact display provides visual feedback so that a user can see the gesture they performed was properly recognized, while audio feedback allows for the best possible interaction between users and non-sign language users. The addition of contextual processing, through sentence-level interpretation and optional text recognition within the environment, is to make the flow of conversation more natural.

V. PROPOSED METHODOLOGY

A. System Architecture

The general operating architecture of the suggested EdgeSign system, which mixes edge and cloud processing, is shown in Fig. 3. At the edge level, the Raspberry Pi utilizes the camera to catch hand gestures and extracts hand markers using MediaPipe. Using WebSocket communication, these processed landmark values are then sent to the cloud. Machine learning algorithms identify the gesture in the cloud layer and create pertinent text that is further transformed into speech. This approach enables the system to react rapidly at the edge while yet using the sophisticated processing power of the cloud.

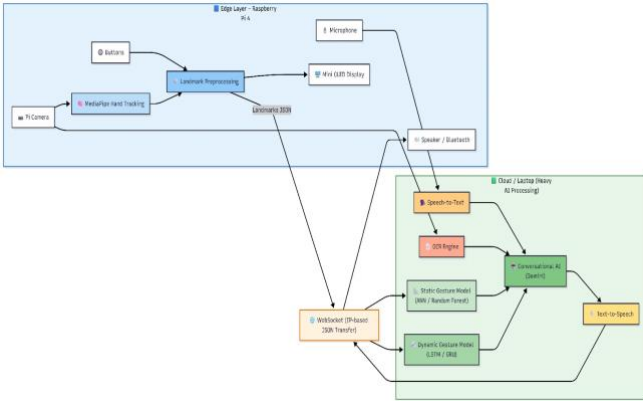


Fig. 3. Architecture diagram

Through a WebSocket connection, the gathered gesture data is delivered to a laptop or cloud based processing unit. This kind of processing unit is capable of computationally demanding jobs including finding the created environment and dynamic gesture categorization. Once the gestures interpretation is over, the generated text is returned to the Raspberry Pi, where a small OLED screen displays it and a text to speech engine turns it into speech. This design enables natural and consistent communication while allocating computing load between the cloud and the edge.

B. Functional Modules

1. Gesture Data Acquisition: The pi camera which record hand motions in real time is used to accomplish gesture acquisition. Two operational modes are supported by the

system the static gesture mode and dynamic gesture mode. In static mode a single frame is taken when the user steadily maintains a gesture in dynamic mode a series of frames is recorded across a set time period to catch movement based gestures. mediapipe gives a organized numerical representation appropriate for machine learning processing by 21 hand landmarks for each frame.

2. Preprocessing and Feature Representation: Prior to training, the square hand landmark data acquired using MediaPipe goes through a process of pre-processing for uniformity and stability. Zero values have been used for missing landmark values (not detected) to create a consistently sized feature vector. The organized (structured) coordinates of these landmarks are laid out so that each hand can be independently analyzed in 3D. To improve classification performance and Z-score normalization was applied by performing standard feature scaling to equally contribute. A hybrid representation of the features was taken by using both raw 3D landmarks and geometric features such as the distances from the wrist to each fingertip, the hand's width, the hand's center centroid, orientation vectors and the distance from one hand to another.

3. Gesture Classification: A Random Forest based model was used to classify both static & dynamic gesture types based on hand landmarks. For dynamic gestures, landmark values are collected from multiple images in order to contain motion information as opposed to using recurrent or sequence neural networks. Based on these classifications, we use the gesture type label as an input to other components that process language at a higher level.

4. Sentence Generation and Context Processing: The system uses AI to create a meaningful sentence from a series of recognized gesture labels (rather than creating a word-for-word translation). In addition, the AI uses an algorithm to help determine when the order of gestures should affect the meaning of the final sentence, thus allowing for a grammatically coherent and natural sentence to be produced from the gesture labels used to create the sentence. The result is a clearer method of communication than any existing stand-alone systems for translating gestures into written form.

5. Output and User Interaction: Attached to the Raspberry Pi a mini OLED screen shows the ultimate interpreted text. The same time a text to speech module transforms the text into audible speech that can be played via bluetooth or wired speaks. Physical buttons are used so that userd may erase the last word or whole sentence.

C. Implementation

Python is used to implement the whole system here MediaPipe forms the main framework for hand landmark identification. Real time data transfer between the Raspberry Pi and the cloud processing unit is accomplished using WebSocket communication. The modular design lets every component gesture capture, preprocessing, categorization, sentence generating, and output run independently, thereby facilitating debugging and future improvements. The

suggested approach uses cloud-based intelligence with effective edge processing to create precise, low latency, and scalable gesture to speech communication appropriate for actual world implementation. Operable continuously utilizing a portable power supply, the system guarantees reliability over extended use. Furthermore the lightweight approach of data transfer preserves user privacy and lowers network overhead. This framework for implementation also makes simple integration of upcoming upgrades including larger gesture sets and better language modeling possible.

VI. RESULTS AND METRICS

Response time, recognition accuracy and usability were among the factors considered when assessing the new gesture to speech device. When the project was tested under regular interior lighting the system with the help of a Pi Camera and a Raspberry Pi 4. Several user's static and dynamic hand gesture motions were taken as the input. Every gesture was identified in real time during the interaction with the system, therefore demonstrating that the output that was received was very reliable and sensitive.

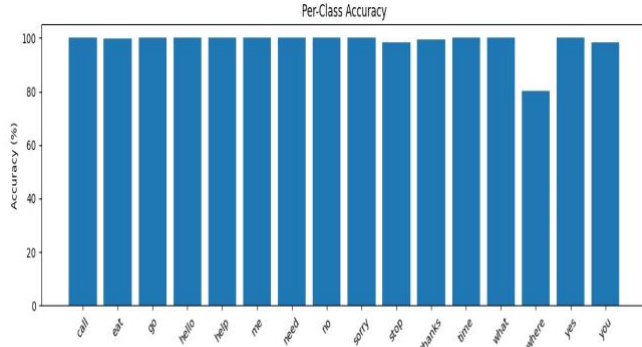


Fig 4. Per Class Accuracy of Labels

Fig. 4 shows how accurate the trained Random Forest gesture recognition model is per class and emphasizes consistently good performance over all specified gesture categories. The majority of courses have precision ratings near to 100%, therefore showing strong model classification ability and efficient feature extraction via MediaPipe hand landmarks. Although overall class wise performance is still quite consistent, minute differences seen in a few gesture categories point to small overlaps in gesture patterns or variability in the dataset. Fig. 5 shows the entire classification report together with the confusion matrix, where high precision, recall, and F1-scores are noted for almost every gesture type, yielding a total test accuracy of around 98%. The confusion matrix shows little misclassification among gesture classes, therefore supporting the claim that the suggested model separates among gestures with great generalizing ability and stability.

TABLE 1. Accuracy Table

Accuracy: 0.98				
Label	Precision	Recall	F1 Score	Support
call	1.00	1.00	1.00	208
eat	1.00	0.98	0.99	289
go	1.00	1.00	1.00	196

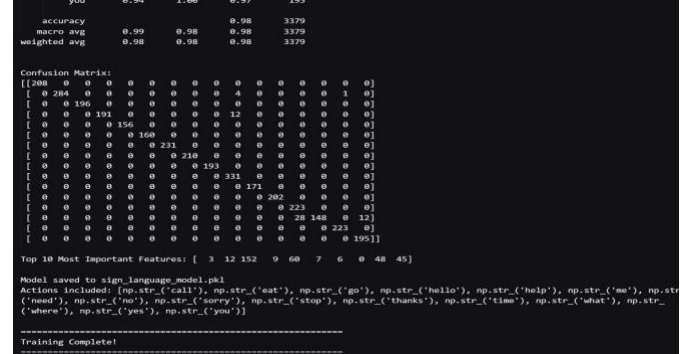


Fig. 5. Confusion Matrix showing correct and incorrect classification distribution

Apart from gesture recognition the system makes sure reliable contact by means of effective edge cloud coordination. Through hand landmark extraction on the Raspberry Pi and strict landmarks data rather than raw video frames and the system conserves network traffic while preserving user privacy

VII. CONCLUSION AND FUTURE SCOPE

This work presented a real time gesture to speech assistive communication system which aims at improving accessibility for individuals with speech and hearing impairments. The proposed system utilizes a Raspberry Pi 4 and MediaPipe-based hand landmark extraction to recognize both static and dynamic gestures efficiently. By using an edge cloud hybrid architecture and transmitting only landmark data instead of raw video, the system achieves low latency, reduced bandwidth usage, and enhanced privacy. The integration of sentence level interpretation, text display, and speech output enables more natural and easy communication compared to conventional gesture-to-word translation systems.

Future improvements in systems should expand the gesture vocabulary of existing systems so that users can express themselves more using different types of sign languages as well as regional dialects. Text-to-speech systems with support for multiple language, hence the goal of providing increased accessibility using multiple language sources. One other way to expand the system will be through the addition of body pose and facial expression data, which will allow the system to gather more contextual information. Improving the performance of these systems through additional on-device processing will also assist in the development of offline functionality and decrease the reliance on remote processing.

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